# An anisotropic Gaussian counterexample to the observable-gradient coupling conjecture

MathDB problem 339682

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**Task 1.** Resolve the claimed sign of the variance change for couplings built from the observable gradient. This is Conjecture 62, equation (157), in the final publication of Nüsken and Pavliotis [1]. The arXiv version numbers it Conjecture 60.

For the overdamped generator $L = -\nabla V \cdot \nabla + \Delta$, the source requires centered $f, \phi \in L^2(\pi)$ satisfying $-L\phi = f$, where $\pi(dx) = Z^{-1}e^{-V(x)}dx$. Its conjectured inequality is

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \nabla\phi(y)^T \alpha_f(x,y) \nabla\phi(x) e^{-V(x)-V(y)}\, dx\, dy \le 0. \tag{1}$$

Writing $\widehat{\nabla f} = \nabla f/|\nabla f|$, equation (156) of the source sets

$$\alpha_f(x,y) = I - 2\,\frac{w(x,y)w(x,y)^T}{|w(x,y)|^2}, \qquad w(x,y) = \widehat{\nabla f}(x) + \widehat{\nabla f}(y),$$

when both gradients and their normalized sum are nonzero; otherwise it sets $\alpha_f = I$. Admissibility in equation (151) is $\alpha_f^T \alpha_f \le I$.

**Proposition 1.** *Inequality* (1) *is false for the smooth Gaussian potential*

$$V(x) = \tfrac{1}{2}(x_1^2 + 9x_2^2), \qquad f(x) = x_1 + 3x_2, \qquad d = 2.$$

*Its left-hand side equals* $56\pi^2/405 > 0$.

*Proof.* Here $Z = 2\pi/3$ and $\pi$ is the centered Gaussian with covariance $\mathrm{diag}(1, 1/9)$. Put

$$\phi(x) = x_1 + \tfrac{1}{3}x_2, \qquad v = \nabla f = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad a = \nabla\phi = \begin{pmatrix} 1 \\ 1/3 \end{pmatrix}.$$

Both linear functions are centered and square-integrable under $\pi$. Since their second derivatives vanish,

$$-L\phi = x_1\partial_1\phi + 9x_2\partial_2\phi - \Delta\phi = x_1 + 3x_2 = f.$$

The observable gradient is the same nonzero vector at every point, so $w = 2v/\sqrt{10} \ne 0$. Thus the nondegenerate branch of the source's definition applies everywhere and gives

$$\alpha_f = I - \frac{2vv^T}{v^T v} = \frac{1}{5}\begin{pmatrix} 4 & -3 \\ -3 & -4 \end{pmatrix}.$$

This constant matrix is measurable and satisfies $\alpha_f^T\alpha_f = I$. The integrand apart from its Gaussian weight is therefore the constant

$$a^T\alpha_f a = |a|^2 - \frac{2(a^Tv)^2}{|v|^2} = \frac{10}{9} - \frac{8}{10} = \frac{14}{45} > 0.$$

Multiplication by $Z^2 = 4\pi^2/9$ gives $56\pi^2/405 > 0$, as claimed. If the integral is expressed using the probability measure $\pi \otimes \pi$, its value is $14/45$; either normalization contradicts the proposed sign. □

The example also defines genuine small perturbations of independent sampling. For $0 < \varepsilon < 1$, the constant diffusion matrix

$$Q_\varepsilon = \begin{pmatrix} I & \varepsilon\alpha_f \\ \varepsilon\alpha_f^T & I \end{pmatrix}$$

has eigenvalues $1 - \varepsilon$ and $1 + \varepsilon$, each twice. With drift matrix $D = \operatorname{diag}(1, 9, 1, 9)$, the process $dU_t = -DU_t\,dt + \sqrt{2Q_\varepsilon}\,dW_t$ has exactly the required two overdamped marginal processes. Its covariance converges to $2\int_0^\infty e^{-Ds}Q_\varepsilon e^{-Ds}\,ds$, which is positive definite, and its mean converges to zero. It is therefore the usual nondegenerate ergodic Gaussian coupling. The counterexample does not depend on using only a possibly degenerate full-strength endpoint.

The source's motivating integration-by-parts identity remains valid: $\int \nabla f \cdot \nabla\phi\,d\pi = 2 = \int f^2\,d\pi$. That identity alone does not force the reflection built from $\nabla f$ to have a nonpositive quadratic form on $\nabla\phi$. Anisotropy of the potential separates these two constant directions. The argument resolves the universal conjecture by a counterexample; it does not decide its restrictions to one dimension or to isotropic targets.

**Task 2.** Determine the sharp condition-number threshold for nonpositive variance change for every linear observable under a Gaussian target, and classify the extremal directions.

**Task 3.** Decide whether the observable-gradient sign conjecture holds in one dimension under smooth uniform convexity of the potential, including observables whose derivatives change sign.

**Task 4.** Construct practical preconditioners for the observable gradient and prove quantitative conditions under which the resulting reflection guarantees a variance improvement.

**Task 5.** Quantify the largest variance increase, relative to independent sampling, for finite coupling strengths in Gaussian targets with prescribed covariance condition number.

# References


[1] N. Nüsken and G. A. Pavliotis, *Constructing Sampling Schemes via Coupling: Markov Semigroups and Optimal Transport*, SIAM/ASA Journal on Uncertainty Quantification **7**(1) (2019), 324–382. doi:10.1137/18M119896X. Section 5.2, pp. 364–365, equations (150)–(157). Final author-hosted publisher PDF.